SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 2.2

Program	**: B. Tech (CSE)**
Specialization	**:**
Course Title	**: AI Assisted coding**
Course Code	**:**
Semester	**II**
Academic Session	**: 2025-2026**
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Enrollment No.	**: 2403A51L53**
Batch No.	**52**
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Submission Starts here :

AI LAB – 2.2

Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI

Task 1: Cleaning Sensor Data

Scenario:

You are cleaning IoT sensor data where negative values are invalid.

Task:

Use Gemini in Colab to generate a function that filters out all negative numbers from a list.

Code Snippet:

```
def remove_negative_values(data):
    cleaned_data = []
    for value in data:
        if value >= 0:
            cleaned_data.append(value)
    return cleaned_data

sensor_data = [12, -5, 7, -3, 25, 0, -10]
print("Before Cleaning:", sensor_data)
print("After Cleaning:", remove_negative_values(sensor_data))
```

Output:

```
Before Cleaning: [12, -5, 7, -3, 25, 0, -10]
After Cleaning: [12, 7, 25, 0]
```

Task 2: String Character Analysis

Scenario:
You are building a text-analysis feature.

Task:
Use Gemini to generate a Python function that counts vowels, consonants, and digits in a string.

Code Snippet:

```python
def analyze_string(text):
    vowels = "aeiouAEIOU"
    vowel_count = 0
    consonant_count = 0
    digit_count = 0

    for char in text:
        if char.isdigit():
            digit_count += 1
        elif char.isalpha():
            if char in vowels:
                vowel_count += 1
            else:
                consonant_count += 1

    return vowel_count, consonant_count, digit_count

text = "Hello123World"
result = analyze_string(text)

print("Input String:", text)
print("Vowels:", result[0])
print("Consonants:", result[1])
print("Digits:", result[2])
```

How can I install Pythor

Output:

```
Input String: Hello123World
Vowels: 3
Consonants: 7
Digits: 3
```

Task 3: Palindrome Check – Tool Comparison

Scenario:
You must decide which AI tool is clearer for string logic.

Task:
Generate a palindrome-checking function using Gemini and Copilot, then compare the results.

Code Snippet:

```python
def is_palindrome_copilot(text):
    text = text.lower()
    left = 0
    right = len(text) - 1

    while left < right:
        if text[left] != text[right]:
            return False
        left += 1
        right -= 1

    return True
```

```python
def is_palindrome_gemini(text):
    text = text.lower().replace(" ", "")
    return text == text[::-1]
```

Output:

```
Input: "madam"
Output: True
```

Observations on clarity and structure:

- The **Gemini-generated code** is concise and uses Python slicing, making it easy to read and understand.
- The **Copilot-generated code** uses a loop-based approach, which clearly explains the logic step by step.
- Gemini's solution is more suitable for beginners due to its simplicity.
- Copilot's solution is better for understanding algorithmic thinking and logic flow.

Task 4: Code Explanation Using AI

Scenario:
You are reviewing unfamiliar code written by another developer.

Task:
Ask Gemini to explain a Python function (prime check OR palindrome check) line by line.

Code Snippet:

```python
def is_palindrome(text):
    text = text.lower()
    reversed_text = text[::-1]
    return text == reversed_text
```

AI Explanation:

- def is_palindrome(text):
  This line defines a function named is_palindrome that accepts a string as input.
- text = text.lower()
  Converts all characters in the string to lowercase to avoid case-sensitive mismatches.
- reversed_text = text[::-1]
  Reverses the string using Python slicing.
- return text == reversed_text
  Compares the original string with the reversed string and returns True if they are the same; otherwise, returns False.

Students Comments on understanding:

- The AI explanation is clear and easy to understand.
- I learned how Python slicing is used to reverse a string.
- The step-by-step explanation helped me understand palindrome logic.
- I can now confidently read and explain similar Python programs.